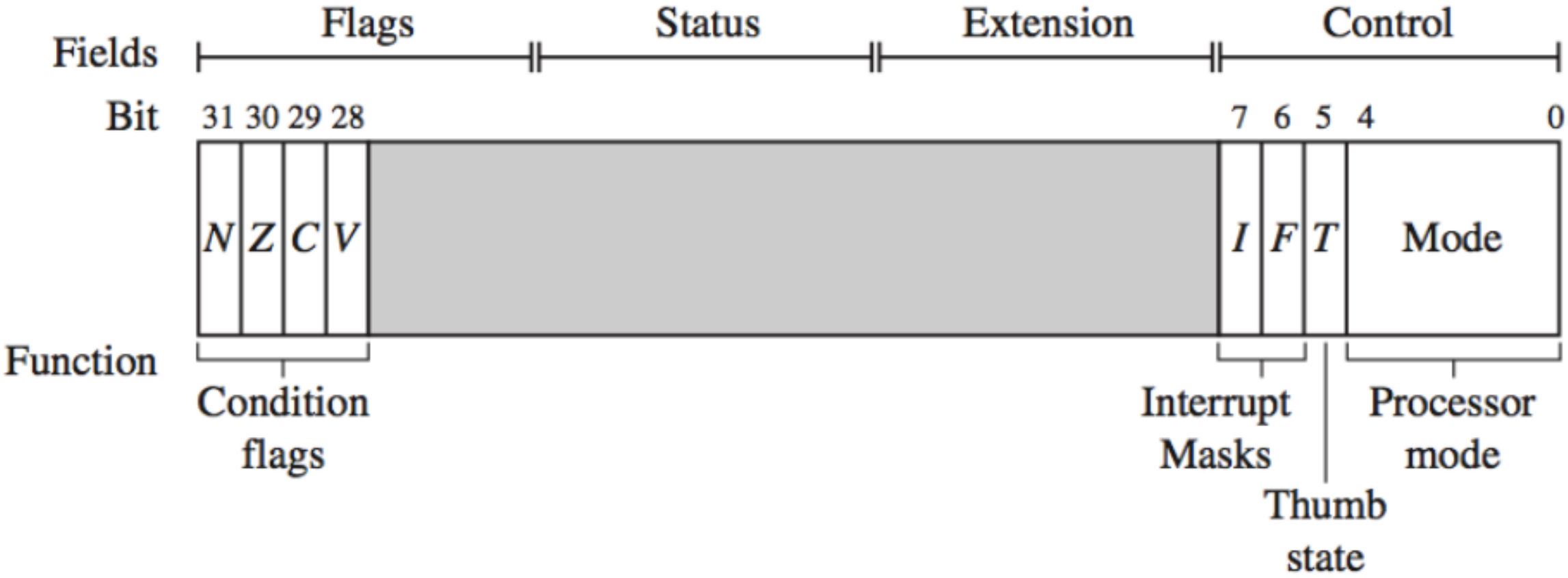
ARM CPSR/SPSR/FLAG

REGISTER



CPSR (Current program status register) holds the current status of the program.
When an exception/ interrupt occurs, the value of CPSR is saved into SPSR (Saved Program Status Register), before invoking the ISR. While returning back to main program, the value of SPSR is put back into CPSR to restore the previous state of the original program.

Mode Bits

ARM7 can operate in 7 different operating modes.
These bits indicate the current operating mode of the processor.

Mode Bits					Operating Mode	Used For
1	0	0	0	0	User (USR)	Normal operating mode
1	0	0	0	1	Fast Interrupt request (FIQ)	When high priority interrupt occurs
1	0	0	1	0	Interrupt Request (IRQ)	When low priority interrupt occurs
1	0	0	1	1	Supervisor (SVC)	Default mode after Reset
1	0	1	1	1	Abort (ABT)	Whenever memory access fails
1	1	0	1	1	Undefined (UND)	When undefined instruction is encountered
1	1	1	1	1	System (SYS)	Can be invoked by programmer

All modes except User mode, are called “Privileged Modes”.

T: Thumb State

If T = 1, then processor is in Thumb state
If T = 0, then processor is in normal ARM7 state.
Thumb state is a special state where "Thumb" instruction set, having 6-bit instructions are used.
This is useful if the memory is implemented as a 16-bit memory.

F: Fast interrupt Mask

If F = 1, then Fast Interrupts are disabled (masked).
If F = 0, then Fast Interrupts are enabled (un-masked).
A fast interrupt occurs through the nFIQ pin.

I: Interrupt Request Mask

If I = 1, then normal Interrupts are disabled (masked).
If I = 0, then normal Interrupts are enabled (un-masked).
A normal interrupt occurs through the nIRQ pin.

Condition Flags

V: Overflow Flag

V = 1 means signed overflow occurred
V = 0 means signed overflow hasn't occurred

C: Carry Flag

C = 1 means carry after MSB

C = 0 means no carry after MSB

Z: Zero Flag

Z = 1 means result is zero.

Z = 0 means result is non-zero.

N: Negative Flag

If N = 1, result is negative

If N = 0, result is positive.

(Note: This flag always indicates the correct sign in spite of an overflow unlike Intel's Sign flag.)